1.05 Linear Modelling

Chapter 5 · Original slide 43 · What's the point of straight line equations?







We saw in Chapter 2 that lots of things in real life have a 'quadratic' relationship, e.g. vertical height with time. Lots of real life variables have a 'linear' relationship, i.e. there is a fixed increase/decrease in one variable each time the other variable goes up by 1 unit.

Examples

Car sales made and take home pay.

(And a pure maths one:)

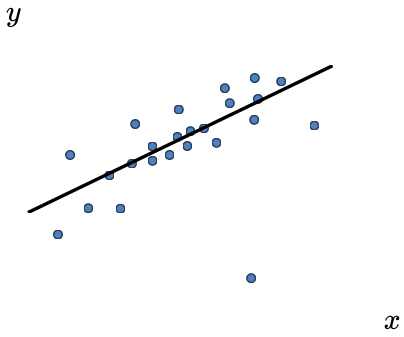
The nth term of an arithmetic series.

3, 5, 8, 11, 14, ...

The relationship between Celsius and Fahrenheit.

Temperature and altitude (in a particular location)

Chapter 5 · Original slide 44 · Modelling



A mathematical model is an attempt to model a real-life situation based on mathematical concepts.

For this example, our model might be a linear model with equation $\mathbf{y=mx+c}$ where x is a student's GCSE mark and y is the predicted A Level mark.

GCSE Maths mark

A Level Maths Mark

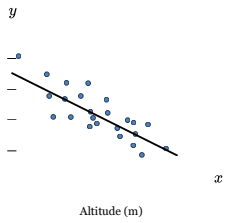
20 40 60 80 100

Such a linear model can be drawn as a line of best fit.

The data obviously doesn't fit this line exactly. This chosen model may only partially fit the data (and the further the points are away from the line, the less suitable this model is).

1.05 Linear Modelling

Chapter 5 · Original slide 45 · Example



100
80
60
40
20

Temperature $^\circ F$

50 100 150 200 250

a b c

The temperature y at different points on a mountain is recorded at different altitudes x.

Suppose we were to use a linear model $y = mx + c$.

Determine m and c (you can assume the line goes through $(0, 70)$ and $(250, 20)$.

Interpret the meaning of m and c in this context.

Predict at what altitude the temperature reaches $0^\circ F$.

Chapter 5 · Original slide 46 · Evaluating if a linear model is most suitable

Choosing a model for our data usually involves making simplifying assumptions. For a linear model we are assuming:

The y value goes up the same amount for each unit increase in x.

Note: Population tends to grow/decline by a percentage each year. Therefore a better model would be an exponential function $P = ab^t$ where a, b are constants. The a would be the initial population and the b the factor increase each year, e.g. $b = 1.1$ would be a 10% annual growth.

The current population of Bickerstonia is 26000. This year (2017) the population increased by 150. Matt decides to model the population P based on the years t after 2017 by the linear model:

$P = mt + c$

Why might this not be a suitable model?

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3 The cost of electricity, E, in pounds and the number of kilowatt hours, h, are shown in the table.

kilowatt hours, h	0	15	40	60	80	110
cost of electricity, E	45	46.8	49.8	52.2	54.6	58.2

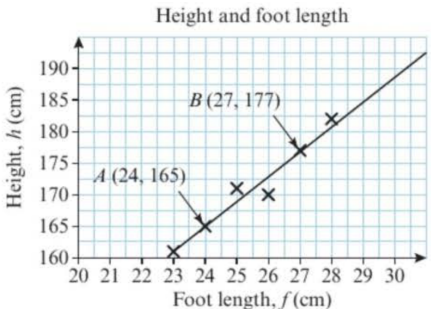
a Draw a graph of the data.
b Explain how you know a linear model would be appropriate.
c Deduce an equation in the form $E = ah + b$.
d Interpret the meaning of the coefficients a and b.
e Use the model to find the cost of 65 kilowatt hours.

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6 The average August temperature in Exeter is 20°C or 68°F. The average January temperature in the same place is 9°C or 48.2°F.
a Write an equation linking Fahrenheit F and Celsius C in the form $F = aC + b$. (3 marks)
b Interpret the values of a and b. (2 marks)
c The highest temperature recorded in the UK was 101.3°F. Calculate this temperature in Celsius.
d For what value is the temperature in Fahrenheit the same as the temperature in Celsius?

1.05 Linear Modelling

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9 The price P of a good and the quantity Q of a good are linked.
The demand for a new pair of trainers can be modelled using the equation $P = -\frac{3}{4}Q + 35$.
The supply of the trainers can be modelled using the equation $P = \frac{2}{3}Q + 1$.
a Draw a sketch showing the demand and supply lines on the same pair of axes.
The equilibrium point is the point where the supply and demand lines meet.
b Find the values of P and Q at the equilibrium point.

8 The scatter graph shows the height h and foot length f of 8 students. A line of best fit is drawn on the scatter graph.
a Explain why the data can be approximated to a linear model. (1 mark)
b Use points A and B on the scatter graph to write a linear equation in the form $h = af + b$. (3 marks)
c Calculate the expected height of a person with a foot length of 26.5 cm. (1 mark)